

CALORIE ESTIMATION USING DEEP-LEARNING FOR DIETARY IMAGES

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ABSTRACT

A balanced diet is a rudimentary and very foremost need for good health, hence the only way to mitigate diseases like diabetes, heart issues or obesity. Obesity is related to many chronic diseases and can be handled with restriction of food intake in addition to your understanding about calories eaten. Recent developments in science and technology have made it possible to create new systems for monitoring, daily consumption of food types such as administration amount in calories from fruit and vegetables. This type of system also gives health information without requiring expert dietitian advice. But with advanced technology that is capable enough to create a system which counts the calories of fruits and vegetables so as to have daily diets. A deep learning model can precisely classify meals sorts and additionally anticipate the calorie content material of unique kinds, assisting human beings keep away from visits to dietitians for ideal health-relevant consumption and dietary plan.

OBJECTIVES

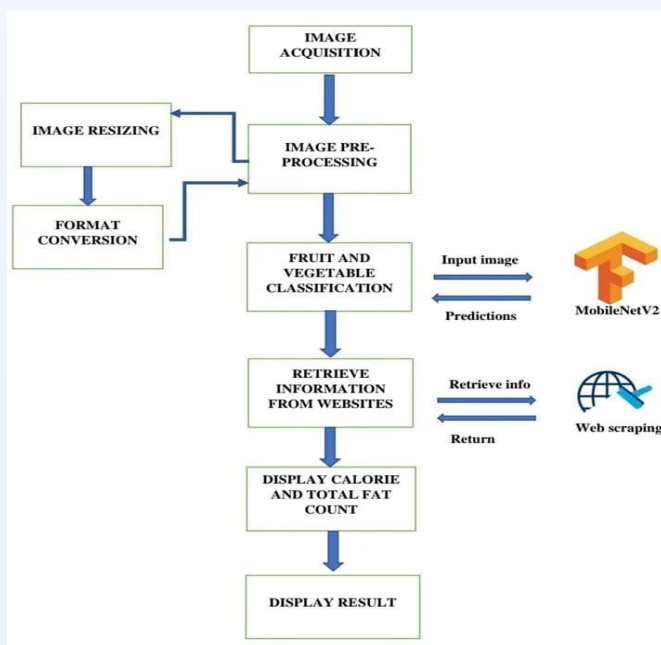
Healthy diet and nutrition are unquestionably necessary for survival, physical development, mental growth, performance and productivity. To maintain good health and for well-being of oneself the following objectives are being defined.

- To propose a framework to estimate Calorie and fat content for different kinds of fruits and vegetables by applying deep learning.
- To identify the variety and healthy food for a dietary.
- To quantify the intake of food for a dietary based on diet plans.
- To make one fit with good health and longevity.
- To develop a functional prototype (web application) for the proposed framework.

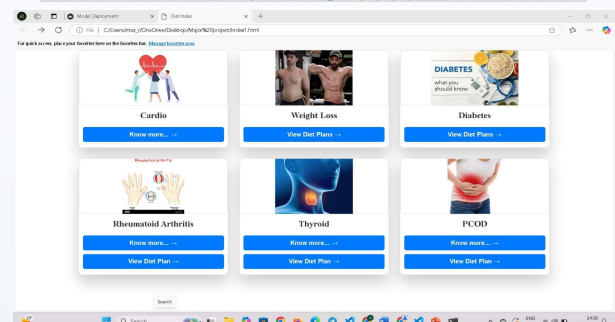
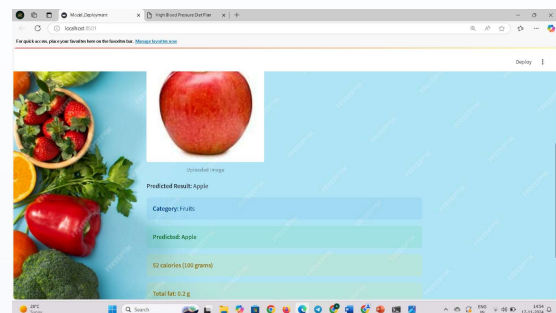
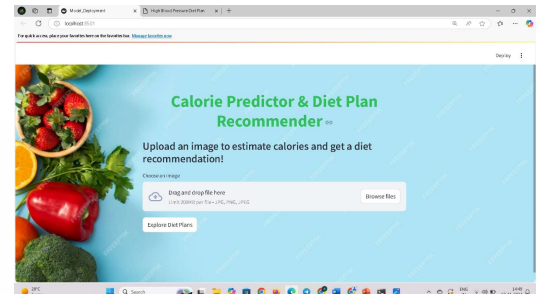
EXISTING METHODS

Author	Title	Year	Method
Manpreetkour Basantsingh	Calorie measurement & classification of fruits using image processing	2016	SGM,SVM
Rutuja Rewane et.al	Food Recognition and Health Monitoring System for Recommending Daily Calorie Intake	2019	F- Shift Invariant Feature Transform, CNN
Ghazanfar Latif et.al	Automatic fruits calories estimation through CNN	2020	CNN
Zihao Zhang et.al	Image Based Food Calories Estimation Using Various Models of Machine Learning	2020	SSD,VGG-16
V Balaji Kasyap et.al	Food Calorie Estimation using Convolutional Neural Network	2021	Random Forest, SVM, CNN

DESIGN ARCHITECTURE



RESULTS



CONCLUSIONS

Uncontrolled and imbalanced calorie intake are key contributors to obesity and related health issues. Many people lack awareness or access to proper diet plans, especially for disease-specific needs. Traditional food logging methods are inefficient and tedious. Leveraging advancements in deep learning and affordable technology, this project proposes a web application to classify fruits and vegetables, estimate their calorie and fat content, and provide tailored weekly diet plans. This tool simplifies dietary management, raises awareness about balanced calorie intake, and educates users on disease-specific diets. By promoting a healthier lifestyle and addressing current health challenges, the project has the potential to evolve into a comprehensive solution as it expands its food recognition capabilities.

REFERENCES

- Cederholm T, Barazzoni R, Austin P, Ballmer P, Biolo G, Bischoff S C. ESPEN guidelines on definitions and terminology of clinical nutrition. Clin Nutr. 2017;36: 49–64.
- Organization WH. Diet, nutrition and the prevention of chronic diseases. World Health Organ Tech Rep Ser. 2003; 916:1–149.
- Organization WH. Diet, nutrition, and the prevention of chronic diseases. Report of a WHO Study Group. World Health Organ Tech Rep Ser. 1990; 797:1–204.